

STUDYING YOUR POOP



ABOUT FAECES, COMPONENTS AND INGESTION

FORMATION



- In the small intestine, food is digested and **nutrients are absorbed**
- **Indigestible materials** (e.g. fibre) enter the large intestine
- **Water is absorbed**, thickening remaining matter
- **Gut bacteria act** on residues and multiply
- Waste is compacted into **faeces** and stored before egestion

IS IT EDIBLE?

- Primarily for **animals**
- Discouraged for **humans**
- Faeces contain **bacteria, viruses and parasites** that are harmful to human body

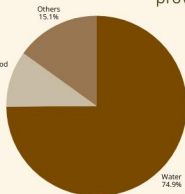
COMPOSITION OF FAECES



OTHERS
A mix of **dead and living bacteria** (gut microbiome), mucus, shed intestinal lining cells, fats, and bile pigments that provide color.

UNDIGESTED FOOD

Comprised mainly of **cellulose** and plant matter that the body **cannot break down**. This **fiber** provides the "bulk" needed to **stimulate intestinal movement**.



WATER (75%)

Provides the necessary **moisture** for **smooth passage**. It is the largest component of **healthy waste**, and its **balance** determines **stool consistency**.

ANIMALS EATING FAECES

- ! **Rabbits**: Redigests their faeces to extract more nutrients from plant based diet
- ! **Hamsters**: Eat their own droppings to get vitamins B and K from gut bacteria
- ! **Pigs**: Eat their own faeces to reprocess nutrients and unabsorbed minerals



HUMAN CONSUMPTION

- Coprophagia: **compulsive consumption** of one's own faeces.
- Associated with other **psychiatric conditions** such as depression.
- Coprophagic humans possess **higher risks of infection, diseases and even death**

OTHER USES FOR FAECAL MATTER

Faeces In Medicine:

- In traditional Chinese medicine, since the 4th century, medications like 五灵脂 (WuLingZhi), 鳩糞 (JiuFen) and 黑冰片 (HeiBingPian) **commonly used** to treat gastrointestinal, nervous system, skin, and gynaecological **diseases**.
- In Western medication, microbiomes have been found to **treat and potentially prevent diseases**. From dementia to cancer, microbiomes prime the **immune system**, promote **healing** and provide **resilience**.

Faeces In Energy Generation:

- Can be **converted into raw biogas** through **anaerobic digestion** by **microorganisms**
- Biogas **upgraded** to RNG, a **source of electricity, heat or fuel**

WORLD TOILET DAY

November 19th Every Year



to raise awareness about **sanitation, digestive health**, and the importance of understanding **natural bodily functions**.

